

Own Works

Composing with Algorithms
<http://www.bjarni-gunnarsson.net>

Dried Up (2008)

Dried Up

Material derived from *synthetic* **percussion** sounds.

Composed *purely* **bottom-up** by arranging the original sounds (and their variants) in clear **gestural phrases**.

Focus on **transformation** for *developing* material and building up sections.

Large-scale form considered as the **result** of combining the phrases and transformations.

Dried Up

Transformations mainly made using the **CDP** system.

Important processes include **waveset distortion** (delete, repeat, multiply/divide) and **extend** (zigzag, random walks, scramble).

Simple **algorithmic processes** applied for choosing different transformations and sequences of treatment.

A **gritty surface** emerges due to the *artefacts* of the various manipulations.

Dried Up (2008)



Bjarni Gunnarsson

Safn 2006-2009

<https://bit.ly/2KYx5ZW>

Portholes (2012)

Portholes

Processes, relations and interaction between elements.

Material defined by its behaviour.

Encapsulation of compositional data, change and possible actions.

Higher level control for accessing different time scales simultaneously.

Sound Processes, Composer Objects and Polarities.

EPOC (Environment for Process and Object Composition)

Portholes

Polarity states as **outside-time**, a *sound process* as **in-time** and an *Object* in-between.

Sound Processes as independent **streams** of sound characterised by their behaviour over time.

Collections of *objects* or other materials as **states**. These can also be seen as *nodes*, *blocks*, *meso-structures* or *sections*. *States* can become **higher-level building blocks** for a composition.

Processes grouped into **sections** or **states**. These groups can be subject to higher level control of **polarities**. They can also become layers or even form vertical relations.

Polarities

Speed
(fast/slow)

Density
(dense/sparse)

Frequency
(hi/low)

Amplitude
(loud/soft)

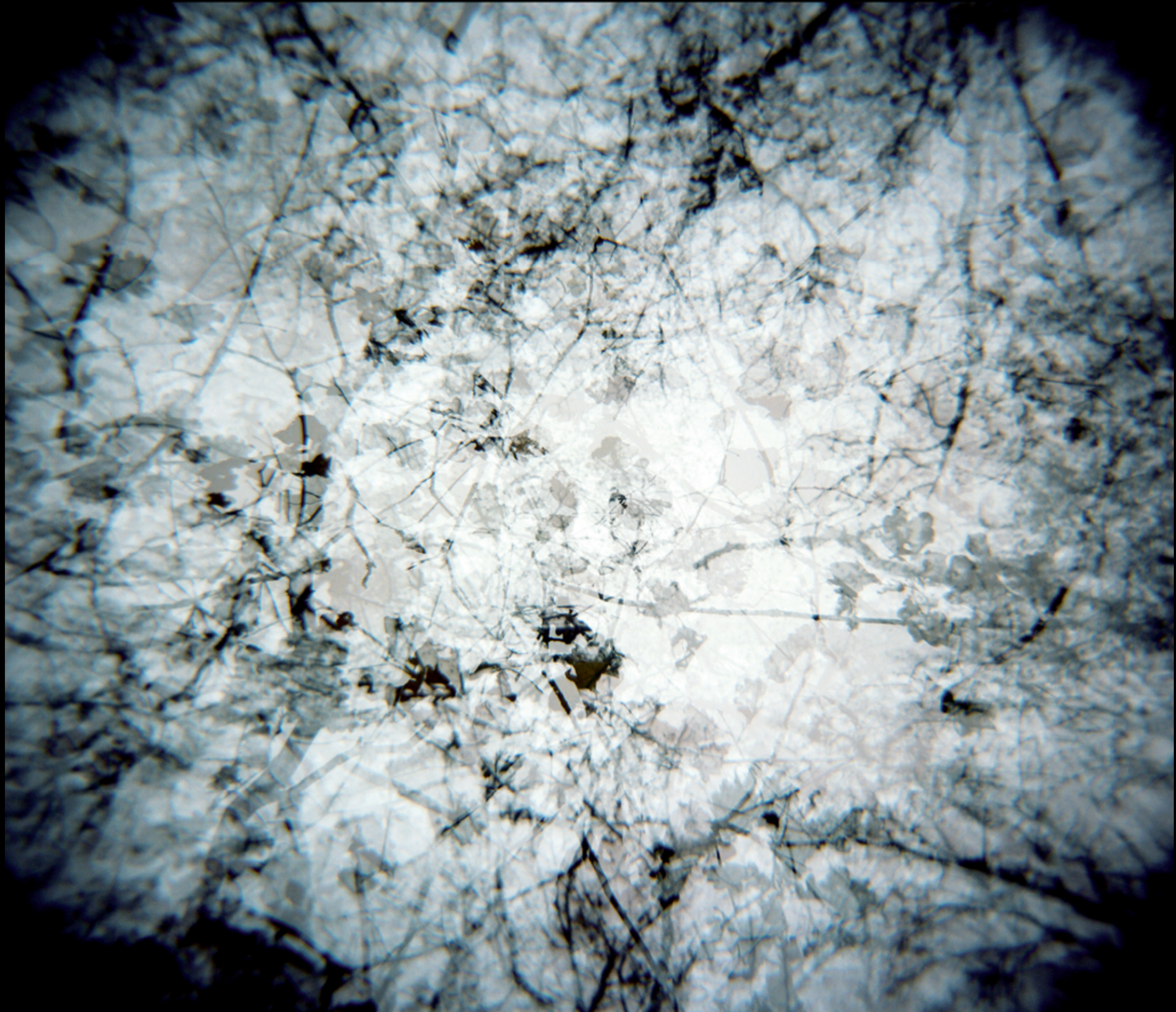
Surface
(rough/smooth)

Color
(dry/wet)

Location
(far/near)

Entropy
(order/disorder)

Portholes (2012)



<https://bit.ly/2wDTUik>

Verlat (2014)

Verlat

Composed for **groups** of sound materials that interact in various ways.

These appear in temporal **states** where the focus passes from one configuration to another.

Snapshots between which the piece rapidly *navigates*.

Switching between states becomes the *driving force* of the music.

The **relation** between states, their **activity**, the time staying in an **instant** (or moving away from it) and the **energy** moving from one *moment* and *transforming* to the next is what the piece is mainly concerned with.

Verlat

The material consists mainly of *particle-based* rhythmic sequences and rich, constant noise that is *filtered*.

The movement of the sequences is *mapped* in different ways to *modulate* the noises and the filters.

Each state/section contains a configuration of 2 sequences and 2 noises that can be heard isolated or together at a given moment.

Certain states *overlap*.

Verlat (2014)

Ubieties (2015)

Ubieties

'In music there often exists an unceasing variation of sound in terms of how it is perceived. Even very static situations create perceptual illusions of development, producing prolonged and continuous transitions.'

Ubieties is based on the idea of a sonic flow where different situations are presented without sharp boundaries. We may think of music as navigating between different moments of continuous change. One chooses a starting point and from there tries to reach other positions. The way a contact between these occurs is however always different and depends on the engagement with previous points of encounter.'

Ubieties

Composed of increasingly **continuous** approaches to transitions compared to previous pieces.

Pulsations as an important characteristic of the material.

These form regular *rhythmic sequences, micro-variations* and contribute to **forward flowing** structural processes throughout the piece.

Contains various areas of sonic activity where **singular characteristics** of certain situations guide the way in which events unfold.

Ubieties (2015)

Mecolico (2016)

Mecolico

Material consists of **synthetic** sounds and **chaotic** sequences.

The synths form clearly **pitched** sounds using *Saw* and *Impulse* generators.

These two broad categories are further divided into eight each based on *variations* and *transformations* of the initial material.

Composed by creating 64 **blocks** (8 blocks for the 8 material variants) before imposing any order in time.

The blocks are **coupled in pairs** and their order in time determined with a flow that is goes from the most *rhythmic* to the most *fluid*.

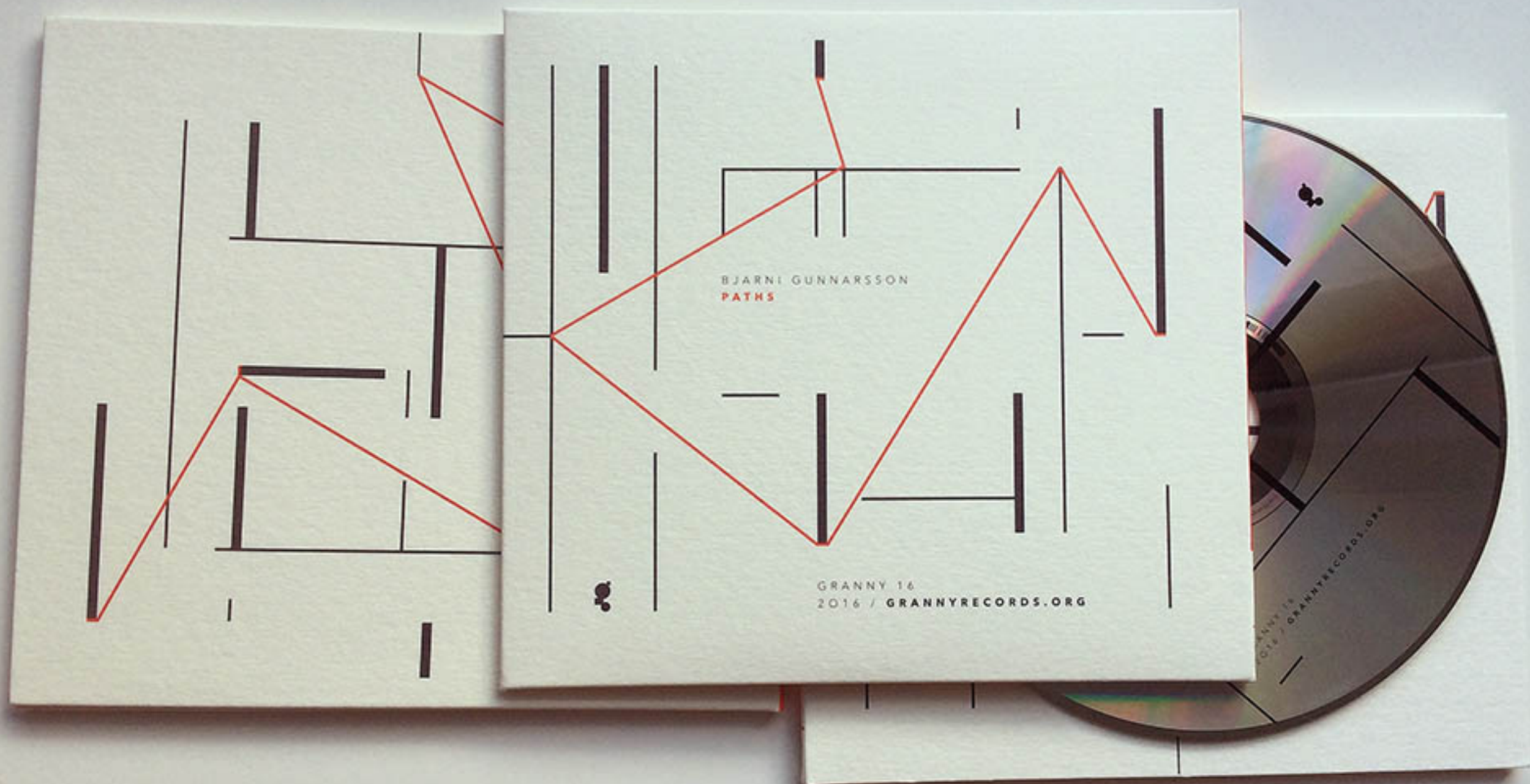
Mecolico

Each of the block has a **parametric representation** for its parts that can be altered to form variations.

Tension between manual edits and algorithmic states.

Spectral **blurring** and **prolongation** are applied to the pitched material that concludes the piece.

Mecolico (2016)



BJARNI GUNNARSSON
PATHS

GRANNY 16
2016 / GRANNYRECORDS.ORG

GRANNY 16
2016 / GRANNYRECORDS.ORG

<https://bit.ly/2KeOqwz>

Prisme (2017)

Prisme

"As processes of light disperse and refract at different angles, broad visual spectra can be produced from pure sources such as white light. The idea of such a multiplication could reflect a possible attitude towards composing sound and the forming of temporal events.

*In '**Prisme**' an attempt is made to gather simple resonances with repetitive rhythmic lines forming multiplicities through its layers, transformations and juxtapositions.*

'Prisme' is part of an ongoing project focused on the use of interactive algorithms for scheduling and organizing synthetic sound sources."

Prisme

Simplification of working method, everything composed through the use of **Pbind** and **SynthDefs**.

Sources in many cases plain synthetics, bells, fm sounds, saw wave glissandis and triangle waves.

Five **sections** lasting from 1:30 to 3:20, similar durations but explore different material and form of layering.

Layers used, three possibilities while defining the piece with the interactive algorithms, other occur during occasional overdubbing.

Prisme (2017)



**BJARNI
GUNNARSSON
— LUEUR**

A. PRISME 12.05
BRACKETS 06.00
B. EPICYCLE 13.05
APERTURE 04.30

Tartaruga Records 2018
Artwork by Luke Twyman
33rpm

15 | 2018

<http://bit.ly/2XvJFG5>

Polytree (2018)

Polytree

Sound processes can be defined by the composition of relations and interconnected parts, where many levels form a greater coherence. Polytree explores how a certain focus on the singular (or local) can introduce movements in time and associations among different elements. Acting on objects that one does not clearly perceive, and their potential to generate new concepts reflects a creative situation in which musical materials are being produced according to their position within a specific context. Instead of autonomy, or isolation, such activity displays a tendency of becoming connected and to be able to adapt. It could be how things originate that gives them their place in time. Perhaps everything is a matter of details, of local conditions and context. Such a contextual based view is an important element of the piece. Polytree is composed in 2018 for an 8-channel immersive sound environment.

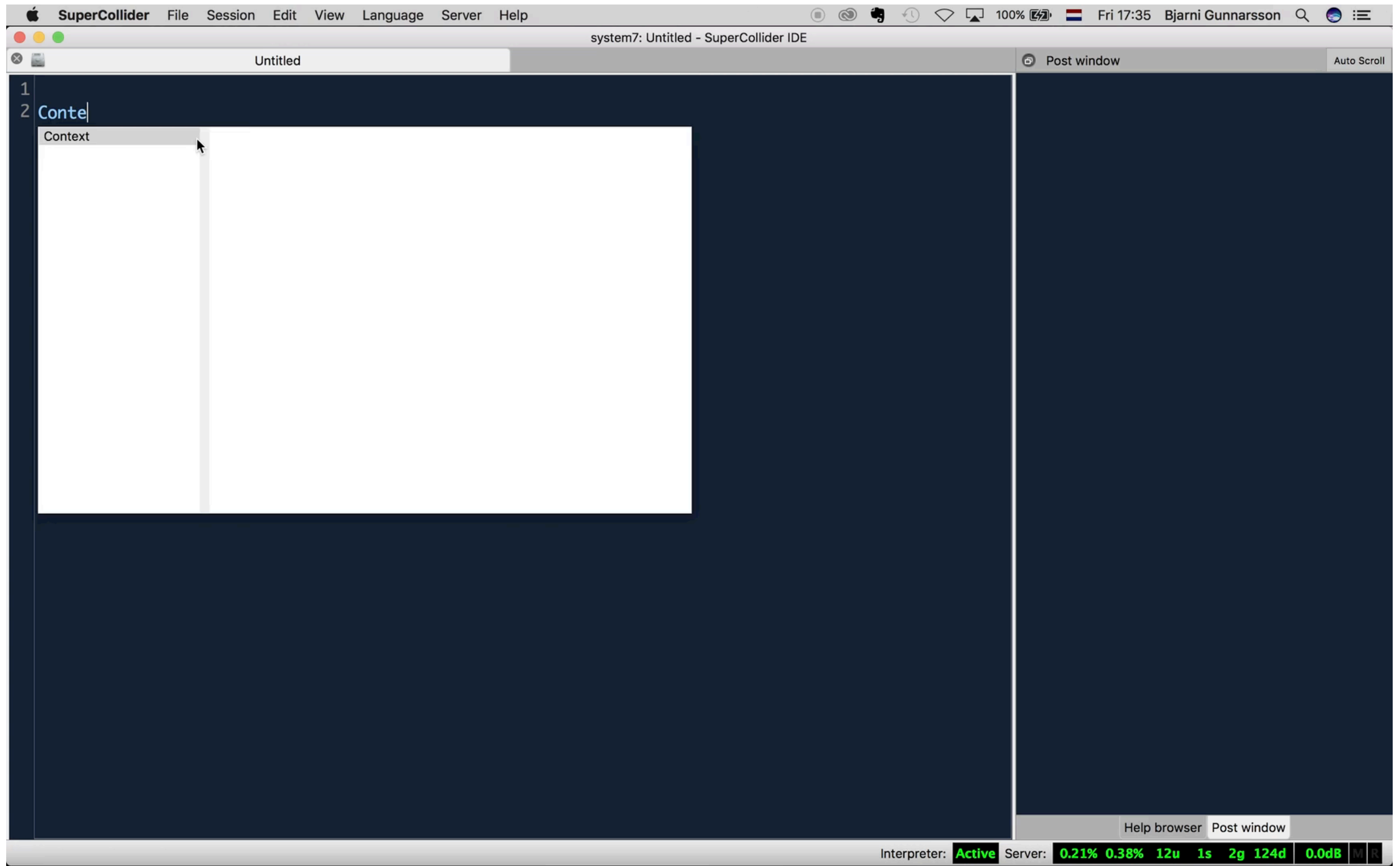
When I direct my attention inward to contemplate my own self [...] I perceive at first, as a crust solidified on the surface, all the perceptions which come to it from the material world. These perceptions are clear, distinct, juxtaposed or juxtaposable one with another; they tend to group themselves into objects [...] But if I draw myself in from the periphery towards the center [...] I find an altogether different thing. There is beneath these sharply cut crystals and this frozen surface, a continuous flux, which is not comparable to any flux I have even seen. There is a succession of states each of which announces that which follows and contains that which precedes it. [...] I could not have said where any one of them finished or where another commenced. In reality no one begins or ends, but all extend into each other.

Henri Bergson

Bjarni Gunnarsson
Cendres



Copaline (2019)



Apple SuperCollider File Session Edit View Language Server Help

system7: seminar.scd (~/Desktop) - SuperCollider IDE

seminar.scd Post window Auto Scroll

```
1
2
3 /* Context.listen → A recording of activity */
4
5 (
6
7 Context.load(this)
8 c = Context(this, "Seminar")
9 c.listen
10
11 )
12
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14 (
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Help browser Post window

Interpreter: Active Server: 0.20% 0.49% 12u 1s 2g 130d 0.0dB M R

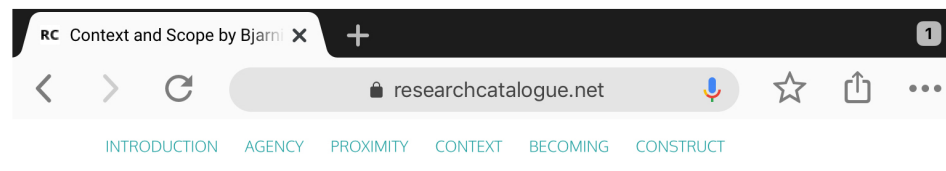
QuickTime Player File Edit View Window Help

system7: seminar.scd (~/Desktop) - SuperCollider IDE

seminar.scd

```
171
172
173
174
175
176
177 /*      Interaction with environment, processes, and persistence      */
178
179 (
180
181 c = Context(this, "Training");
182 c.import()
183
184 )
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```

Interpreter: Active Server: 0.19% 0.35% 12u 1s 2g 138d 0.0dB M R



Context and Scope

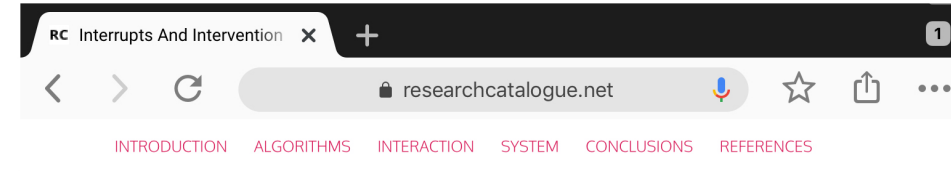
By Bjarni Gunnarsson

This exposition forms part of my research on the positioning of algorithms for composing computer music made at the [The Royal Conservatoire](#) and [Institute of Sonology](#) in The Hague from late 2017 to early 2019. It is motivated by an interest in problems concerning generative algorithms and the apparent environment they form part of. It continues from my previous proposal [considering algorithms as black boxes, accessed from an external point of view and through interrupts and intervention](#).

This text concerns issues of algorithms, agency, proximity, context and the becoming of sound processes. It questions the scope and potential boundaries of an algorithm, the creative space relating compositional practice with computers and the idea of an algorithmic environment. By discussing the properties of context and the boundaries that shape algorithmic processes, the following questions will be addressed:

1. How can an experimental environment be created (or composed) for generative algorithms to operate within, be conditioned by or emerge from?
2. How can interaction with generative processes take place through a model based on an artificial environment or generated context?
3. How can the creation of computer music take place through a dialog with a software environment that is being constructed in parallel to the music that is created with it?
4. How can a generic and abstract compositional algorithm translate to a specific, articulated singularity that is unique to a specific sonic process but also reusable as 'black boxed' operation?

These questions will be addressed by exploring ideas relating to algorithms and their context, by proposing software approaches that address such environments and by discussing the creation of two computer music compositions. [Polytree](#) (2018) and [Copaline](#) (2019).



Interrupts and Intervention

By Bjarni Gunnarsson

Computing today is increasingly oriented towards real-time interaction, networks, distributed systems and adaptive algorithms. Using generative computer processes when creating music raises questions of how to model compositional procedures that allow for openness, interaction and malleable organisation of time. Instead of employing predefined algorithms during composition, processes should be designed to adapt to or evolve within such a situation. The framing of decision-making is essential to the design of such processes.

An important consideration is how to represent a generative algorithm as distributed (open), with multiple entry points instead of being closed and result-oriented only. How to introduce dynamic modifications of goals (or heuristics) as a creative interaction mode or how the attitude of evolving criteria (variable intentions) can be made audible through the characteristics of the corresponding sound processes. The main idea is to separate the scheduling and producing parts of generative algorithms. To balance the influence of processes that take a certain time to manifest themselves and their activation. This involves making the algorithms block-based so that different areas of their evolution can be accessed freely. It also makes use of adaptive heuristics and interrupt-based scheduling.

This exposition forms part of my research on the positioning of algorithms for composing computer music made at the [The Royal Conservatoire](#) and [Institute of Sonology](#) in The Hague from late 2017 to early 2019. Further research on the topic can be found in my work [on the context and scope of generative algorithms within compositional environments](#).

Presented below is a software framework, an interactive, container-based, process scheduler which will be abbreviated in this text as CPSI. It provides an integrated approach to interaction and scheduling of generative processes. CPSI has three main parts: A scheduler, an interaction module and a process container. CPSI can be programmed to run without any user intervention but also to operate within a real-time situation.

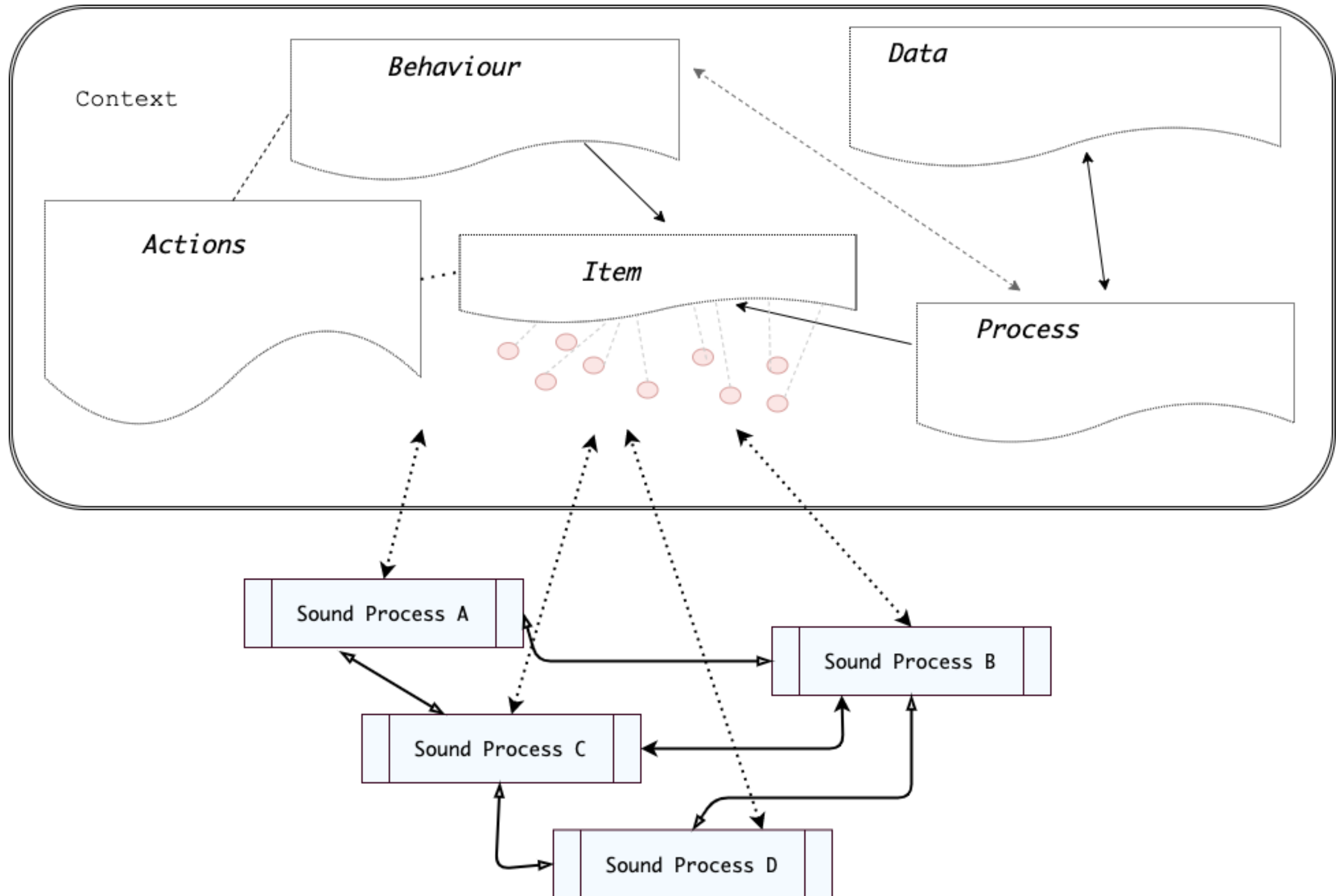
The CPSI framework offers several important functionalities:

<https://www.researchcatalogue.net/view/543155/543156>

<https://www.researchcatalogue.net/view/540350/540351>

<http://www.bjarni-gunnarsson.net>

Volume & Void



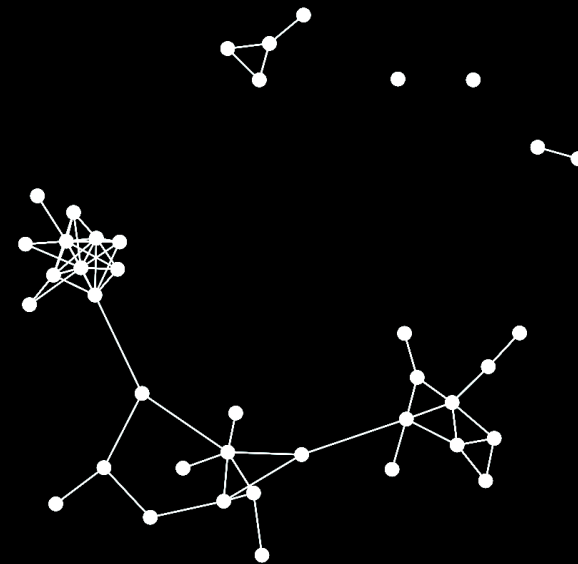
Stash (2020)

BJARNI GUNNARSSON:
VOLUME & VOID 1.HEA
P 2.WRECK 3.BLUR 4.
STASH SYSTEM AND SO
UNDS CREATED DURING
THE SUMMER OF 2020
IN SCHEVENINGEN, TH
E HAGUE 1698000MS

SUPERPANG

SP18

<https://bit.ly/38FSTtX>



Synthesis and Operation Flows

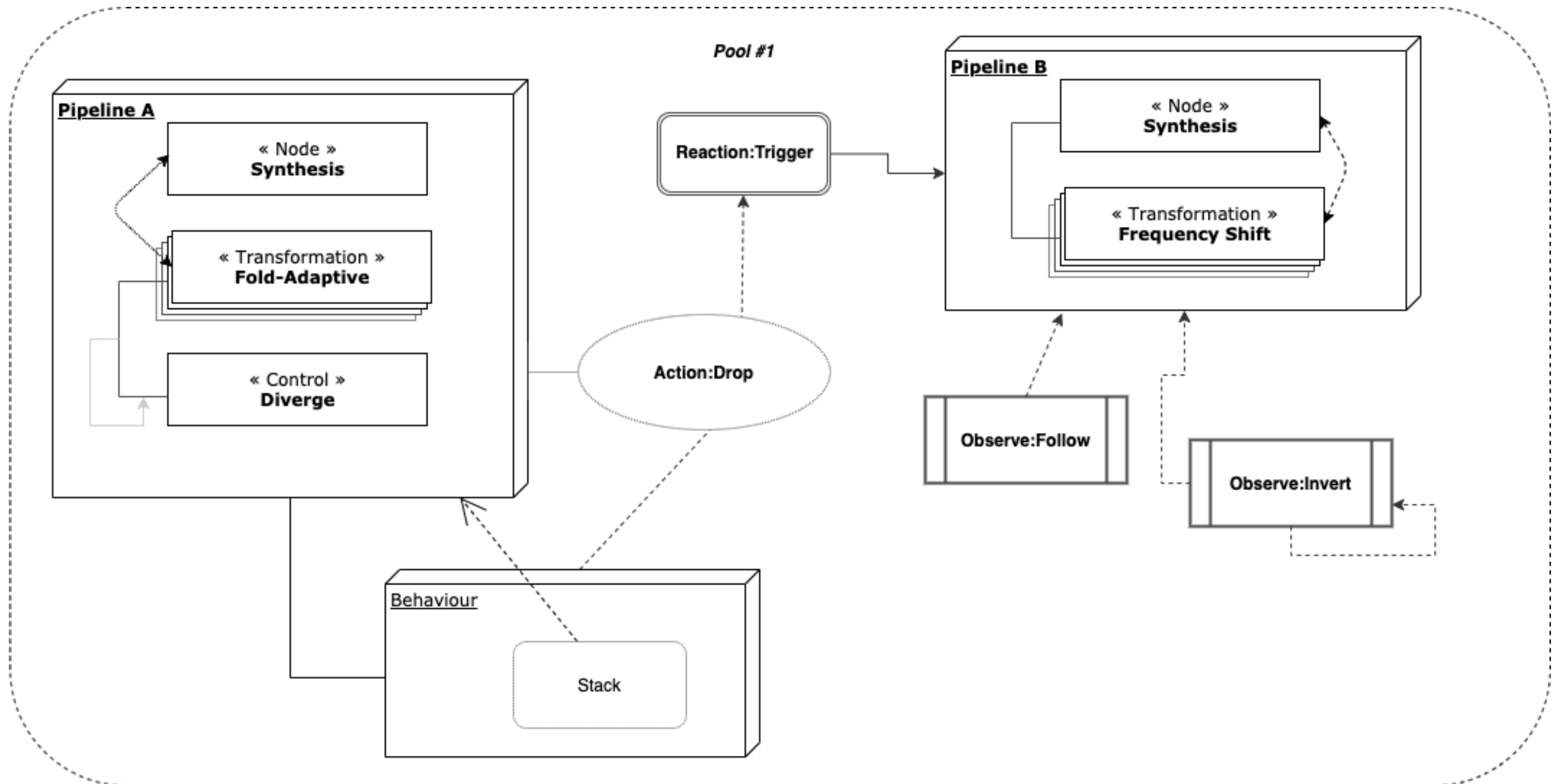
OF

A software framework named OF (Operation Flows) consists of four main process categories:

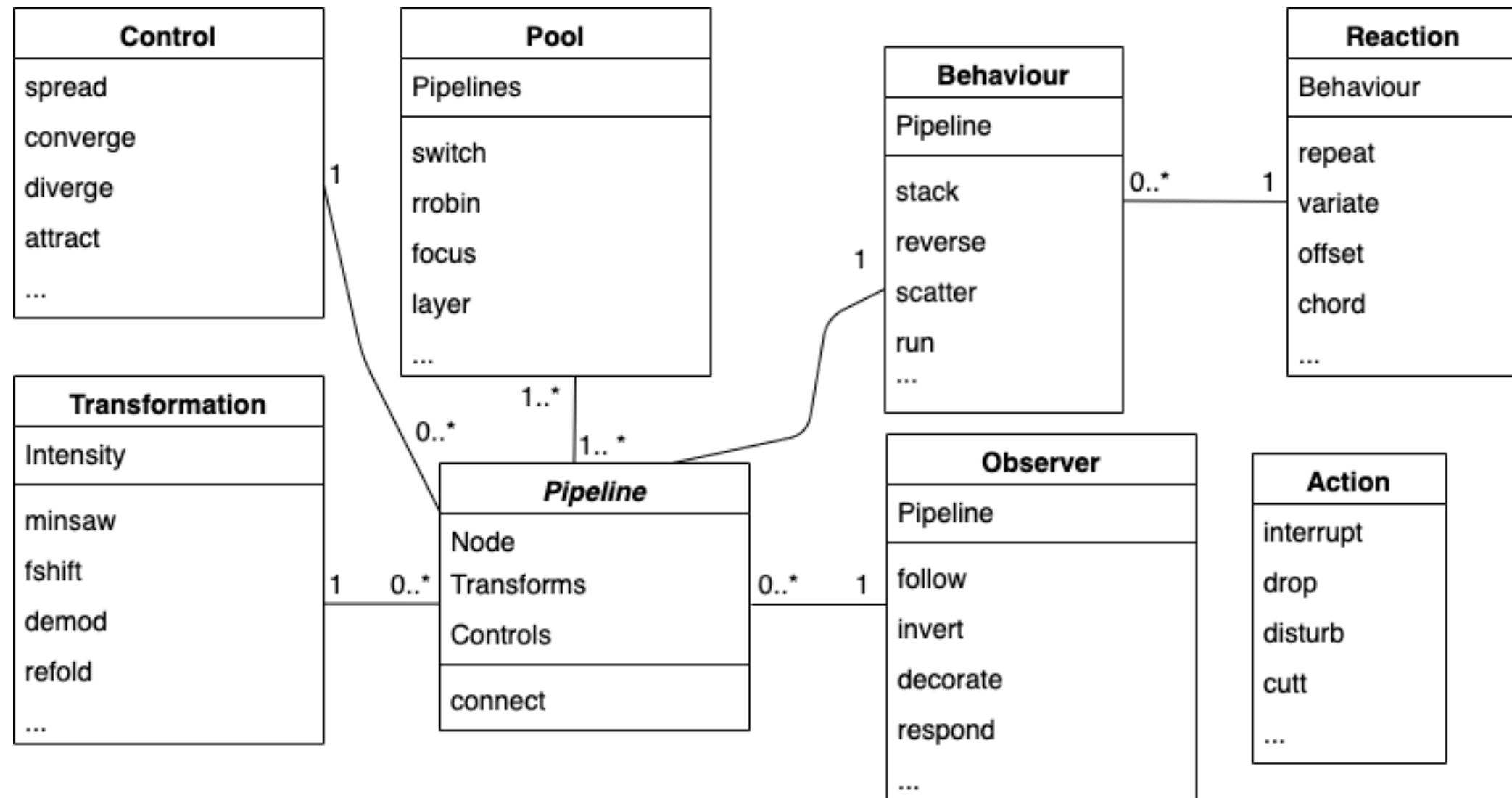
1. **Synthesis processes** with dynamic waveform generation
2. **Atomic sound transformations** that adapt to incoming signals
3. **Operation pipelines** and algorithms to control and generate them
4. **Direct access**, observers and reactions for content-based triggers and actions

The framework allows for experimenting with synthesis and algorithmic pipelines of sound transformations. It is designed to be configured with common existing workflows and to be easily extensible. Examples that demonstrate the relation between the text and the system will be presented here below.

Architecture



Entities



OF - Class Diagram

BJARNI GUNNARSSON ANTICLINES

BJARNI GUNNARSSON

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<https://github.com/bjarnig/Accrete>

Selective Retention

- **Tools and Interfaces**

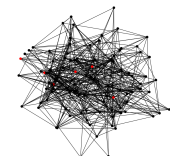
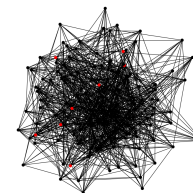
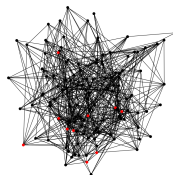
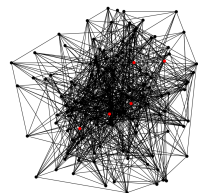
- Tools as **interfaces** to existing material, enabling through **interaction**
- The act of implementation becomes a **method of discovery** and clarification
- Work unfolds through **feedback** between systems, sound, and perception

Emergent Intention

- **Intention** arises within the process rather than preceding it
- Algorithms operate as **acting systems**, shaping both method and result

- **Relational Thinking**

- Composition approached **bottom-up**, emphasizing **material interaction**
- Focus on **local differences** and interdependencies rather than global design
- Creative judgment and **intention emerge dynamically** through making



Relational Systems

- **Behaviour and Interaction**

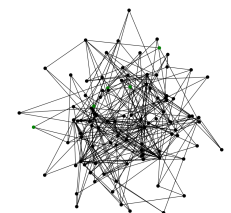
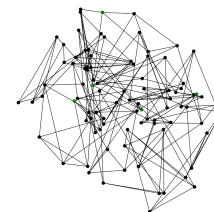
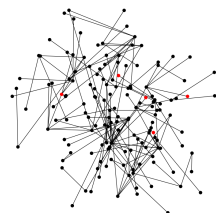
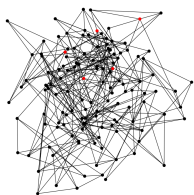
- Sound processes conceived as **behaviours** in continuous feedback loops
- The composer becomes an **observer** of evolving algorithmic relations
- Emphasis on **interaction** over control, where change drives form

- **Network Growth Models**

- **Graph-based structures** explored as compositional frameworks
- **Preferential attachment** for clustering and emergent scales
- Networks treated as **performative configurations**, not fixed designs

- **Causality**

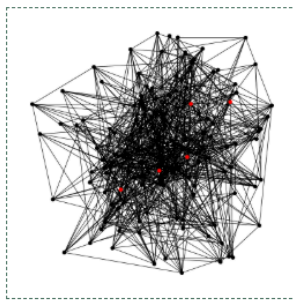
- Events arise through **connections** and **transitions** between nodes
- **Causality** becomes **distributed** and temporal, shaped by sound itself
- Intention as **relational**



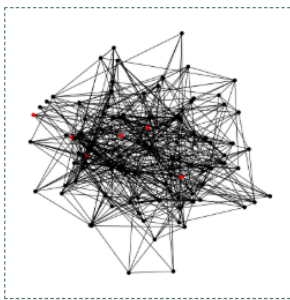
Selective Retention: Interfacing the Past through Queries and Graphs

Bjarni Gunnarsson

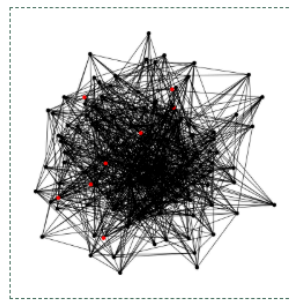
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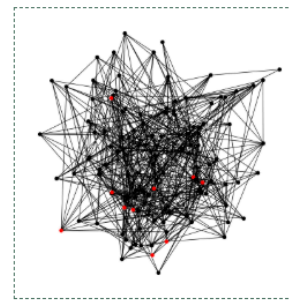
Thought Clarifications



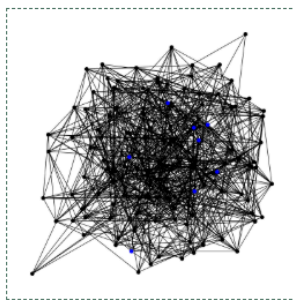
Discovering Intentions



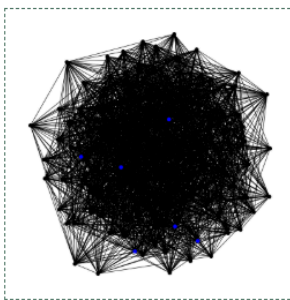
Selective Retention



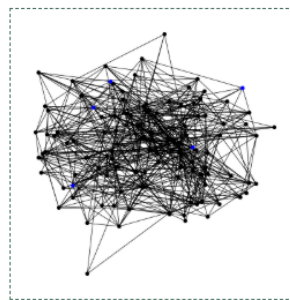
Flow Control



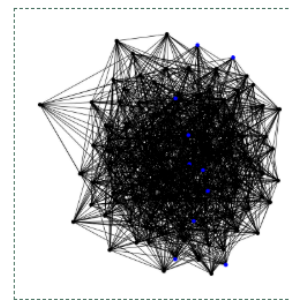
Archived Past



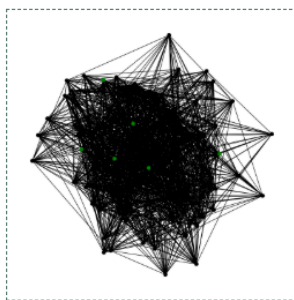
Exploratory Analysis



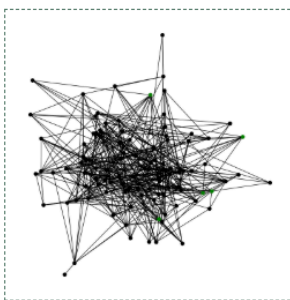
SND Archive



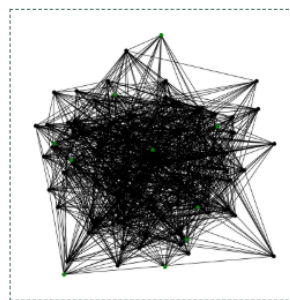
Grouping Discoveries



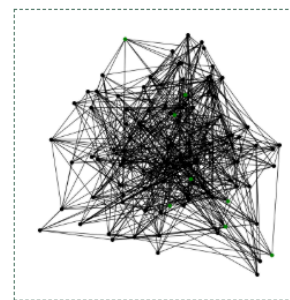
Attaching Behaviours



Framework Sonorities



Preferential Attachment



Holding Pattern

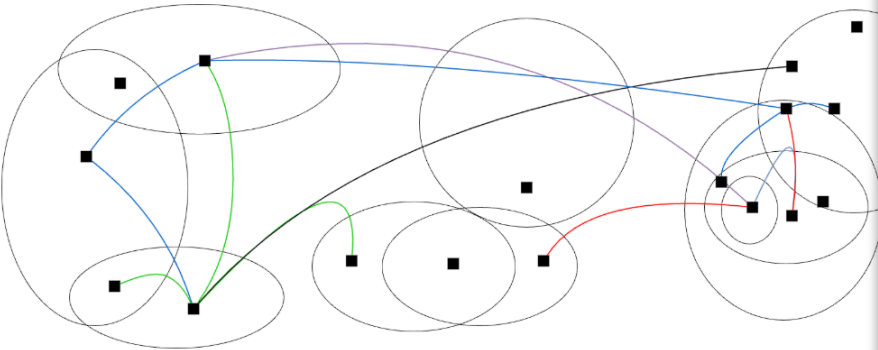
These four files are part of the original sound material that were later used within the systems discussed, and are here included in their original form. The first two from the UPIC system and the second two from the live-coding sessions.



Discovering Intentions

Algorithms are usually thought of as something that has to fit an original intention, goal, or solution. However, the idea of an intention can also apply to something that takes shape during the creative process instead of before it. The (guiding) ‘*intention*’ is then to be discovered and worked out in parallel to the solution it involves. On a similar note, the philosopher Yuk Hui reflects on the objects of computing and the output of technology by arguing that within an artistic context, it is the production of something unknown that should be the goal of the process and the condition for anything to be intelligent within a specific machine. Not only should the unknown be the object of intelligence but also ‘*the condition for the development of such intelligence*’ (Hui 2021: 243). Such applications can be seen as ways of searching, refining, or understanding. The implemented systems (or algorithms) therefore exist as methods of clarification or decoding where system building becomes a productive context for generating knowledge and where algorithms function as ‘*the frontier between thinking and acting*’ (Rohrhuber 2008: 2).

The systems presented here are not understood as static entities or something with a fixed structure. Rather, they come about through their interfaces to existing material and data sets. What matters is the arrangement of their components and the nature of the relationships with the data they process. Such a perspective allows us to work out alterations, elimination, or rewiring of computer programs that take place alongside the development of a research approach and the making of a creative work.



Thought Clarifications	Discovering Intentions	Selective Retention	Flow Control
Archived Past	Exploratory Analysis	SND Archive	Grouping Discoveries
Attaching Behaviours	Framework Sonorities	Preferential Attachment	Holding Pattern

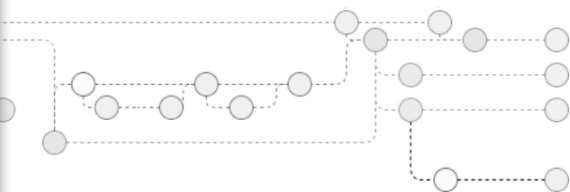
My piece/improvisation 'Fracas' from 2021 is an early example of using a graph-modelling approach to select sound material, as short sequences that appear in networks, where the relationality becomes part of the musical organisation.



Selective Retention

In his book *The Wandering Mind*, Michael C. Corballis reflects on the ‘*default-mode networks*’ of our minds and how the essence of creativity takes place through ‘*blind variation*’, a transition from a set path to something unknown, where what we do stumble across is recognised and labelled as a ‘*selective retention*’ (Corballis 2015). His idea of wandering is not concerned with a specific problem-solving goal but rather with something more uncharted or exploratory. Similarly, composing music can be understood as moving through stages of ongoing transformation. It starts at a certain point, and then ventures along different paths. The relationship between these points can be constantly revised with past points of connection. What stands out are the patterns that emerge at process.

These patterns as domains that can be navigated through brings about new connections between another and what their relations consist of. This allows for the emergence of these connections (and possibility spaces) within the process instead of understanding ‘wandering’ as a way to allocate a path for development, as a selection of choices, as presenting a path that in itself becomes something that provides value. Each path that emerges is an integral part of what is defined, found, and attributed to the process. It turns, and the unique events encountered are what distinguish it with distinct character and creative interest.

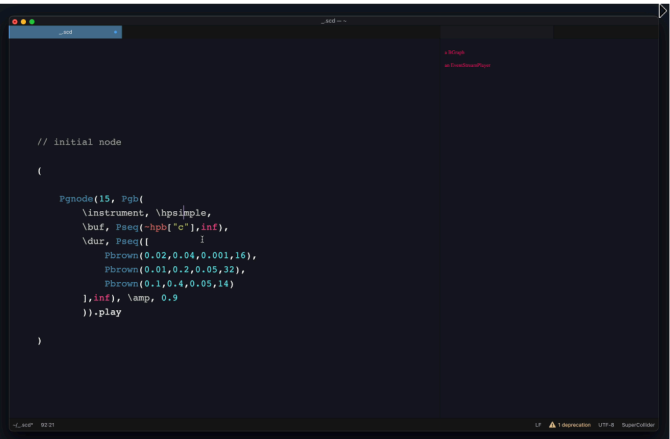


Exploratory Analysis

In my work with digital synthesis, I have become interested in using machine-listening approaches where sound processes auto-regulate based on analysis data. While real-time analysis provides valuable insights, relying heavily on those for crucial behaviours can sometimes result in overly smooth and generalised outcomes. This tendency contrasted with my objective then of achieving sharper and more distinctly articulated results. Using the UPIC material, I wanted to analyse more sounds in depth, compare them, and find gaps and unique perspectives that would be harder with real-time approaches. My methodology was significantly influenced by concepts from data science, especially those introduced by John Tukey in his groundbreaking work ‘*Exploratory Data Analysis*’ (1977). His methodology focuses on discovering the unknown and deeply exploring the inherent qualities of data to build a thorough understanding of a certain dataset under question. His approach emphasises delving into data with curiosity, allowing the data itself to guide the analytical journey and reveal its underlying story.

The UPIC sounds would become my dataset. Could this extensive archive of past sounds serve as a foundational or contextual basis for the development of compositional algorithms? What was interesting about the UPIC archive as a starting point was the strong character of the sound material. Most of the sounds had some sort of pitch movement or otherwise noisier parts. There were very few ‘neutral’ sounds or static textures that would easily combine or melt with similar sounds. There was also a great number of recordings that existed and were organised in the same directory structure as fifteen years before.

This video shows the creation of a graph based on existing patterns that use growth and preferential attachment. A graph is ‘produced’ while playing and can be traversed at a later stage.



Costest

BJARNI GUNNARSSON FRACAS

<http://bit.ly/4w5oefC>

<https://github.com/bjarnig/Ora>

ORA

Ora takes its name from the Latin *ōra*, meaning border, rim, frontier, or edge. The piece unfolds along such boundaries, between sound and process, between tone and texture. All musical material is derived from a small set of sine waves that act as both source and constraint. These sets are stretched, multiplied, and transformed by algorithms, generating textures that continuously reorient themselves. Ora is part of an ongoing project exploring process-based composition through generative design, emphasizing continuity and sustained temporal processes.



Ora

Thank you!

